Group 3

Global Unemployment Market Analysis

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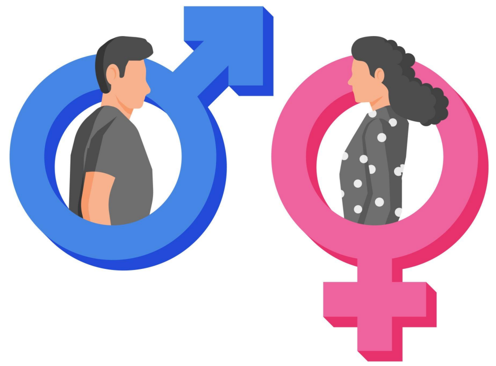
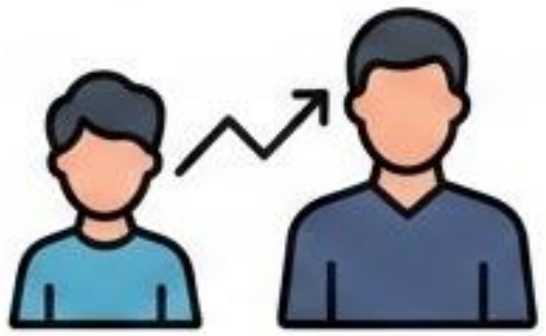
Our Objectives

Macro-Economic Resilience



Structural Gender Parity



Generational Workforce Disparity

By Human Development Index (HDI) | By Education Level | By Continent

Dataset Description

```
Data columns (total 9 columns):
 #   Column                    Non-Null Count   Dtype
---  ------                    --------------   -----
 0   iso_code                  57519 non-null   object
 1   country                   57519 non-null   object
 2   sex                       57519 non-null   object
 3   age                       57519 non-null   object
 4   year                      57519 non-null   int64
 5   Unemployment rate         57519 non-null   float64
 6   continent                 57519 non-null   object
 7   Human Development Index   56889 non-null   object
 8   Education Level           56574 non-null   object
dtypes: float64(1), int64(1), object(7)
```

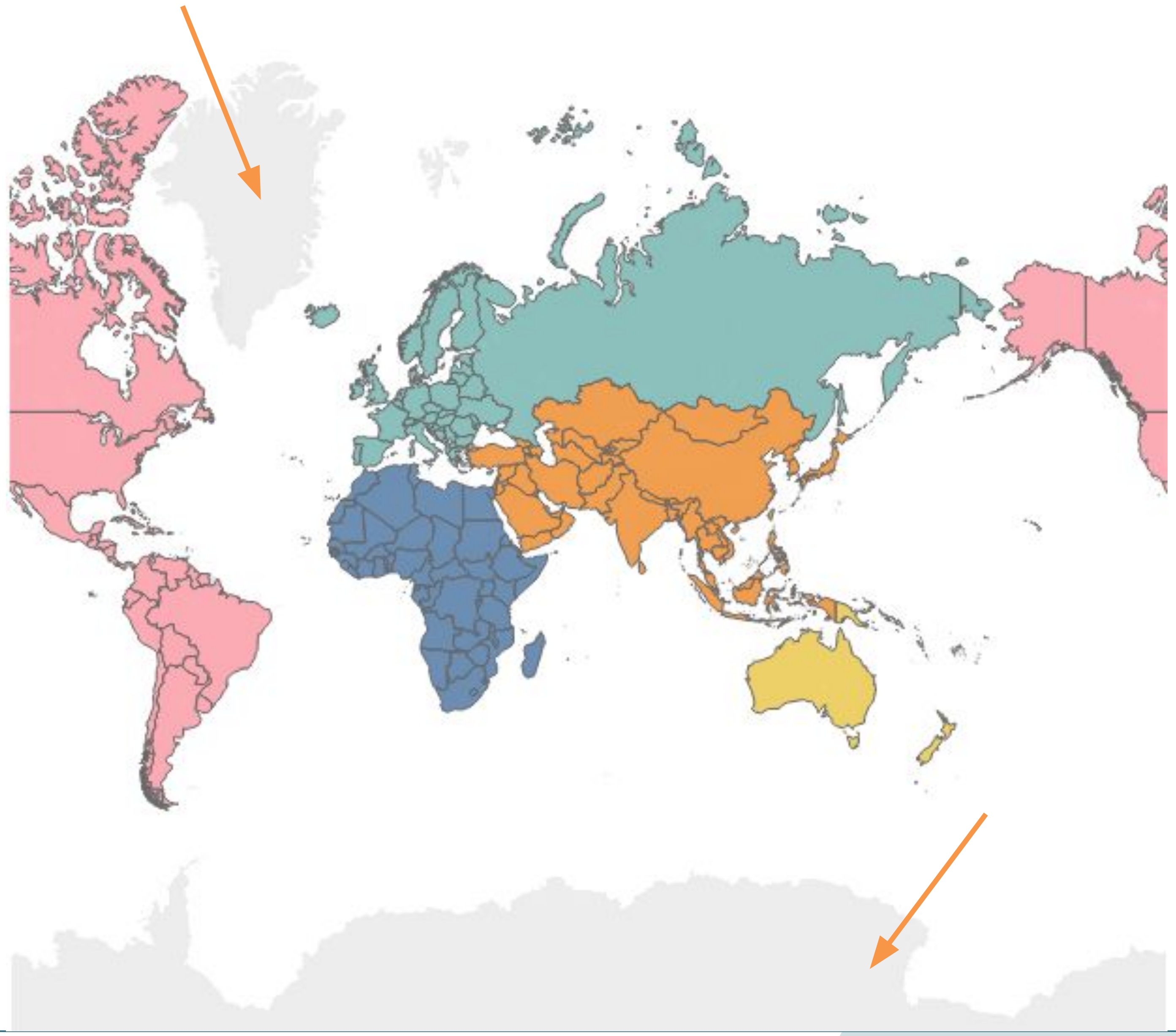
Dataset Expand

Unemployment Rate Dataset

- **183 countries**
- **35 years (1991-2025)**
- **9 data for each country (Sex: M, F, Total, Age: 15-24, 25+, 15+)**
- Data Integrity: ~99.78%
- (Total 126 records not included for 2022-2025)

Countries Not Included

- Greenland
- Antarctica



Dataset Description

Missing Data on Unemployment Rate

5 countries with missing unemployment rate data on 2022-2025

- **Only 2.73%** (5 countries) not included in the dataset
- **Only 2022-2025** are missing throughout 1991-2025

Analyse using Median and Grouping by Continent

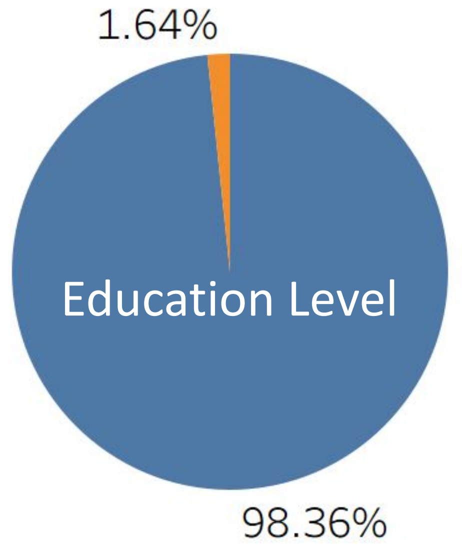
		Year																
Continent	Country	2009	2010	2011	2012	2013	2014	2015	2016	2017	2018	2019	2020	2021	2022	2023	2024	2025
Africa	South Sudan	9	9	9	9	9	9	9	9	9	9	9	9	9	9	9		
	Sudan	9	9	9	9	9	9	9	9	9	9	9	9	9	9			
Asia	Lebanon	9	9	9	9	9	9	9	9	9	9	9	9	9	9	9		
	Palestine (State of)	9	9	9	9	9	9	9	9	9	9	9	9	9	9			
Europe	Ukraine	9	9	9	9	9	9	9	9	9	9	9	9	9				

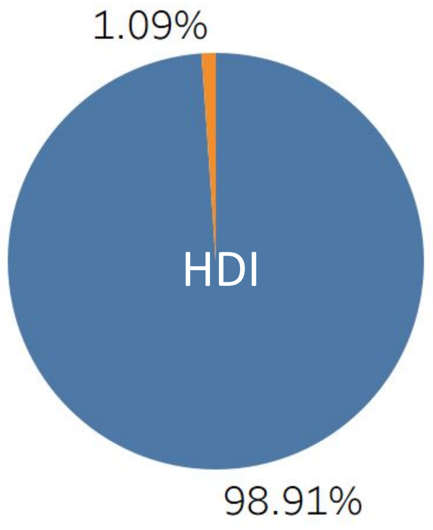
Missing Data on Describing Countries

2-3 countries with missing descriptive information

- Only **1.6%** (3 countries) Missing Education Level
- Only **1.1%** (2 countries) Missing Human Development Index (HDI)

Not alter the overall inference (< 2%)





Our Methodology

Tooling

Tableau Desktop

- Handling Multi-Dimensional Cross-Filtering
- Geospatial Visualization Strength
- Longitudinal Trend Analysis
- Data Granularity & Disaggregation

Visualization Approach

Analytical Approaches

- Geospatial Mapping
- Time-Series Analysis
- Dashboarding

Macro-Economic Resilience

Global Unemployment Trend

By Human Development Index

- Identify Spikes along 1991-2025
- The Impact of Development (HDI)

Highlights

- Cyclical Volatility:
  a. 1998: The Asian Financial Crisis
  b. 2009: The Great Recession (Global Financial Crisis)
  c. 2020: The COVID-19 Pandemic

Highlights Global Unemployment Trend

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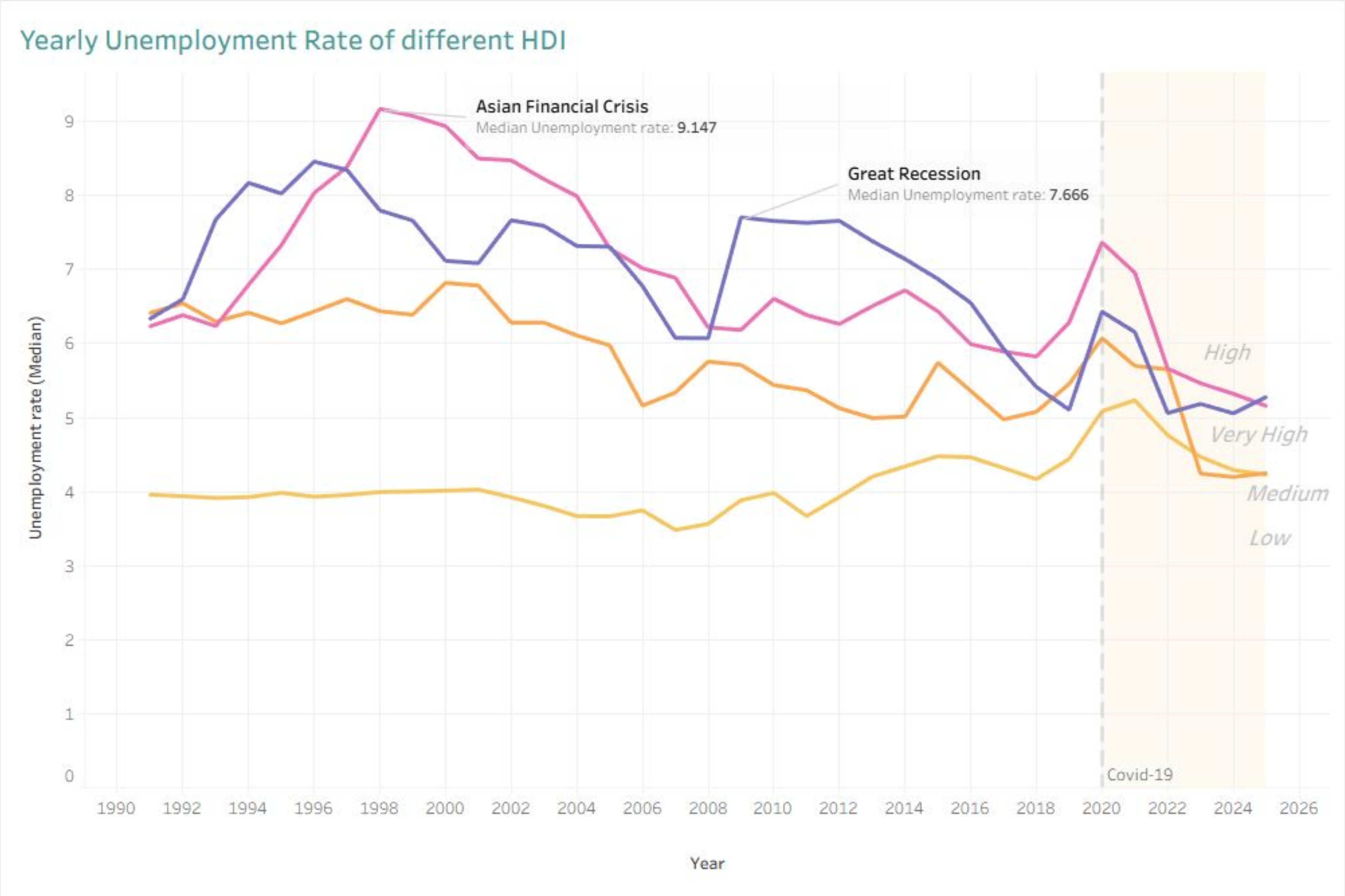


Structural Gender Parity

Gender Inequality

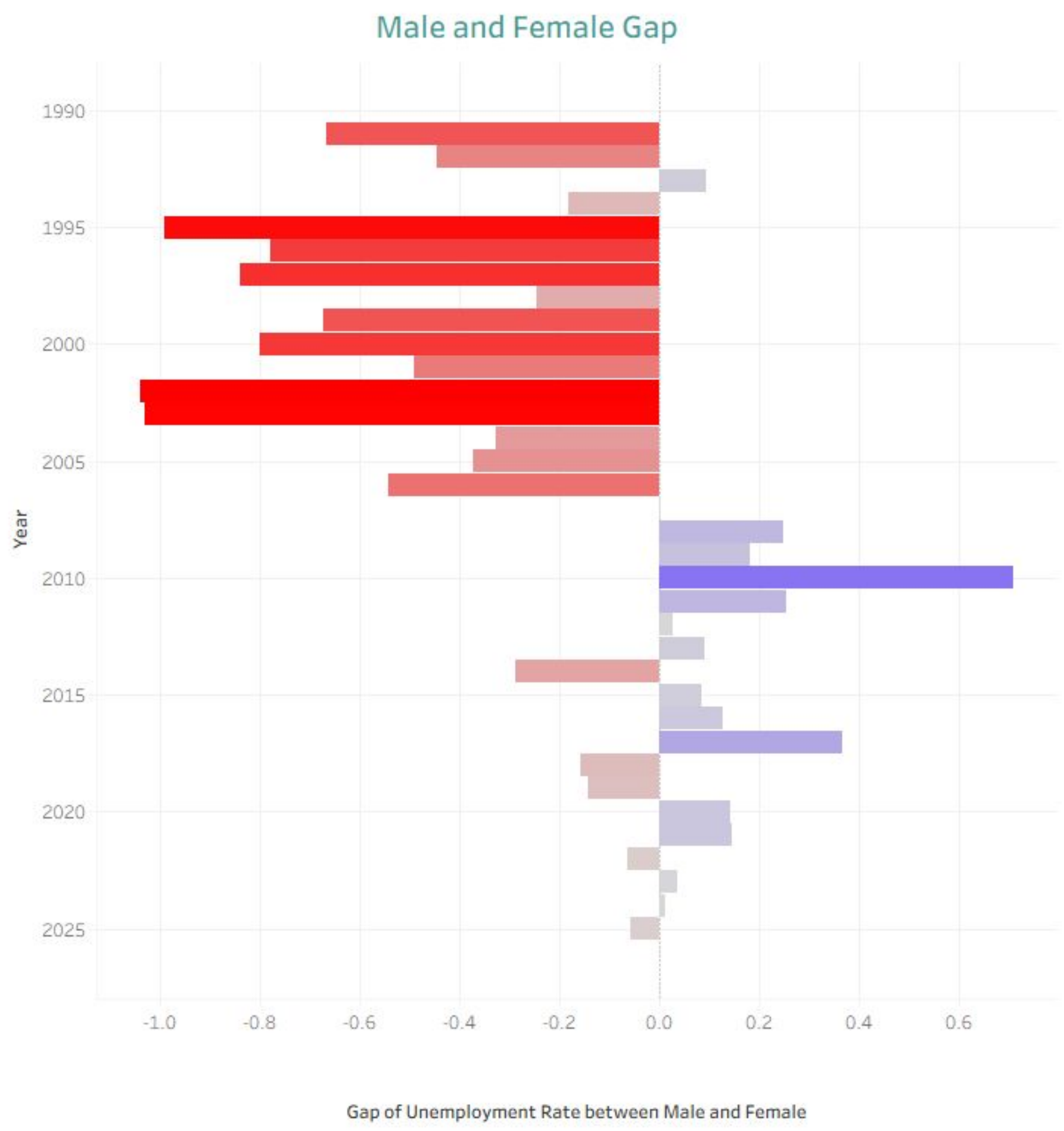
By Education Level

- Gender Gap
- Education as a Balancer

Highlights

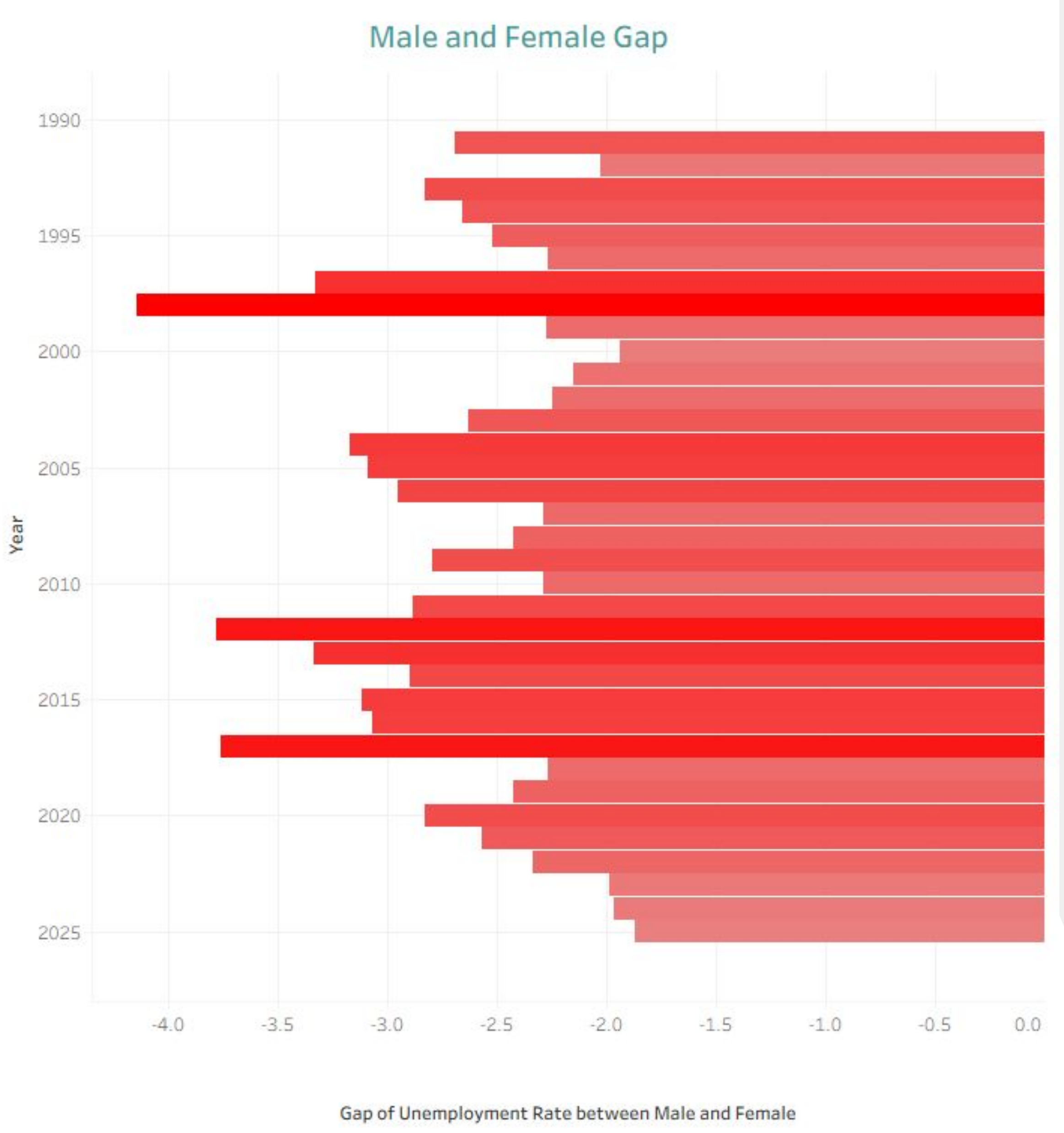
- Female > Male
- Higher educational attainment significantly narrows the unemployment gap

Highlights Gender Inequality



High education level

- Less and less unemployment gap in recent years



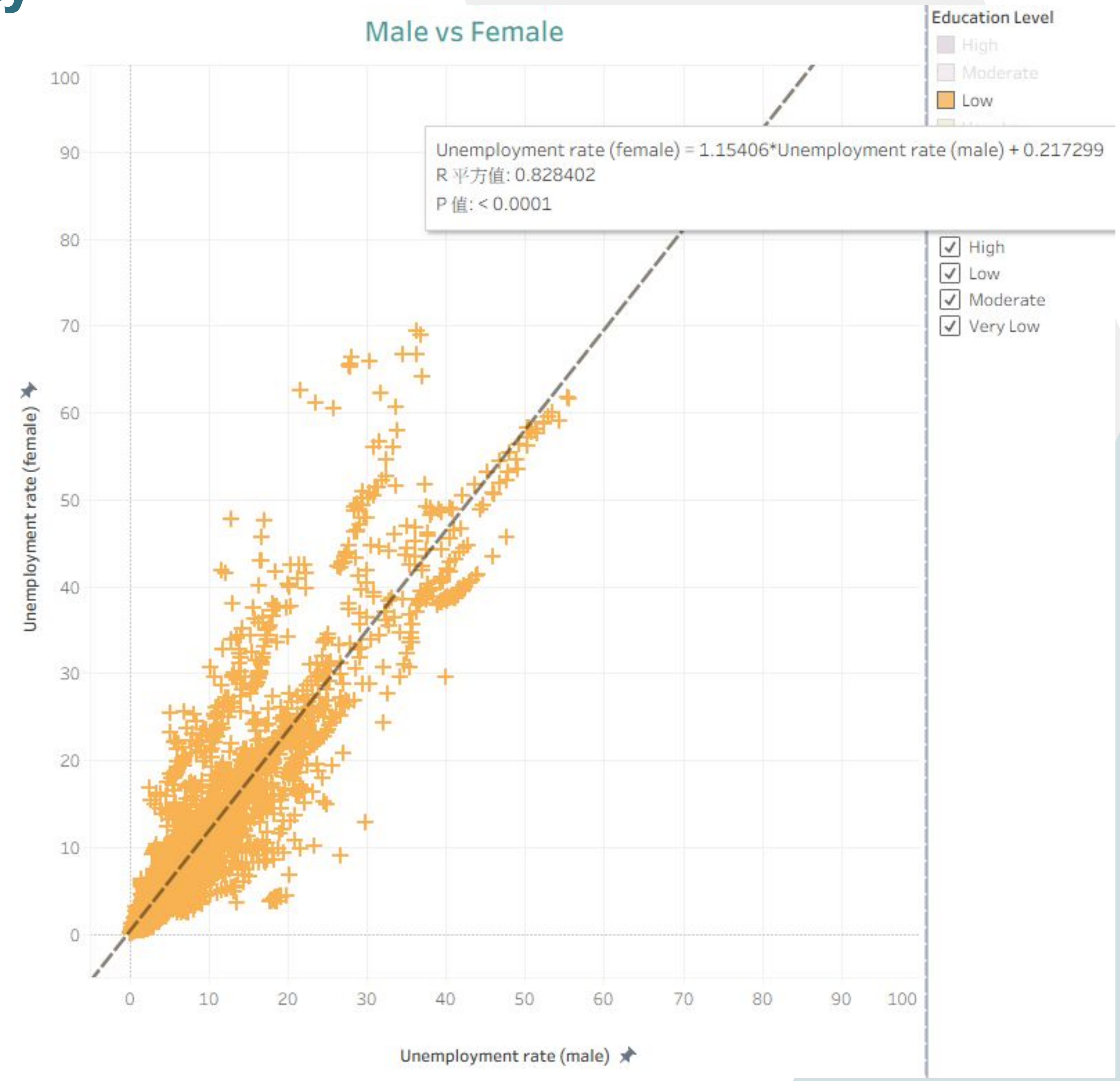
Moderate education level

- Unemployment rate of female higher than male over the years

Highlights Gender Inequality

Low education level

- The trend line has the greatest slope (1.154) among all other education levels
  - More data points on the left side of the graph
  - Female unemployment rate is higher than male



Generational Workforce Disparity

Age Inequality

By Contents

By Continents

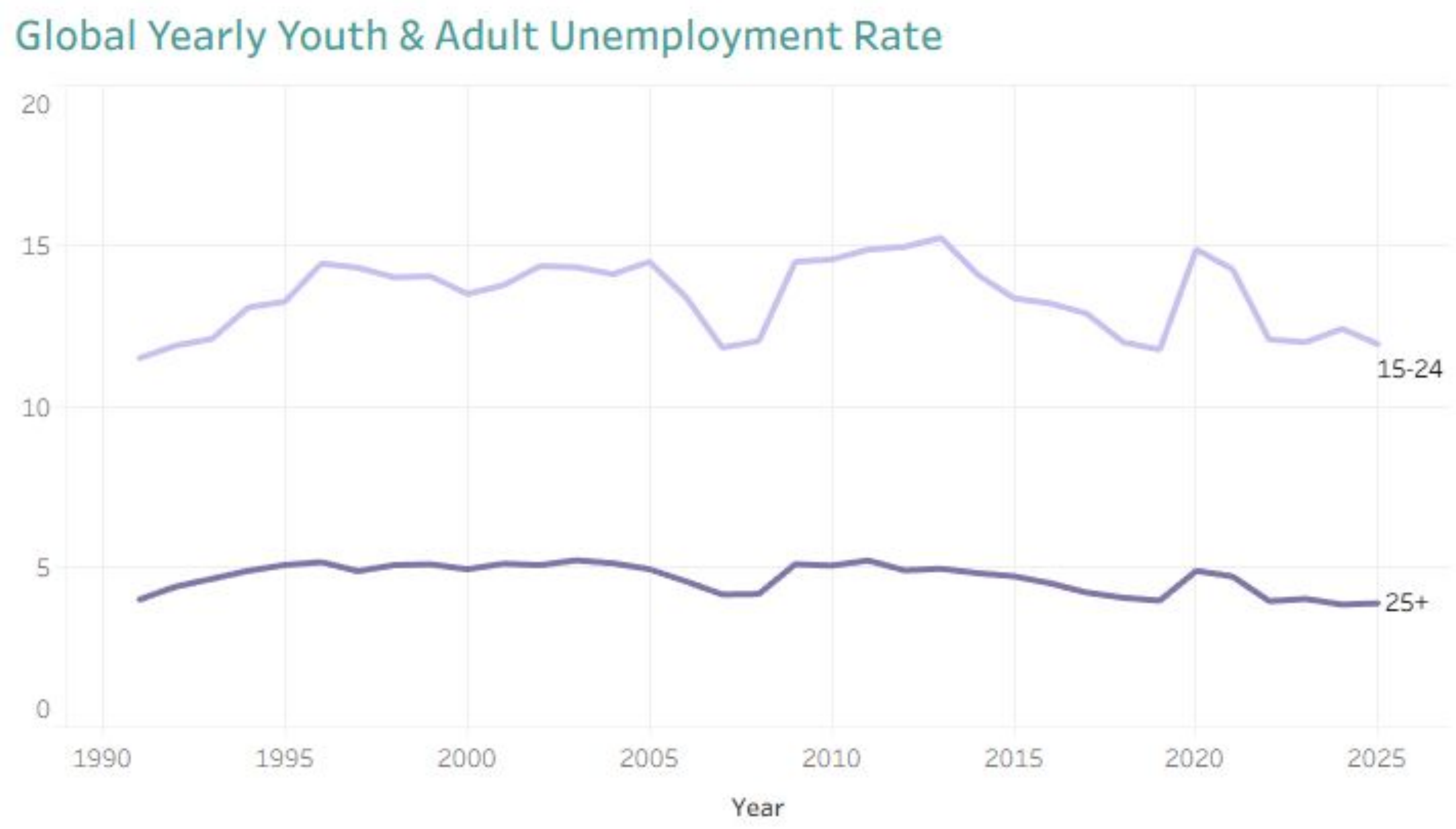
- Economic Sensitivity of Youth and Adult
- Gap between Youth and Adult

Highlights

- Youth obviously higher than adults
- Youth unemployment rate in Europe and America (Higher HDI and education level) is much sensitive and much higher than adult

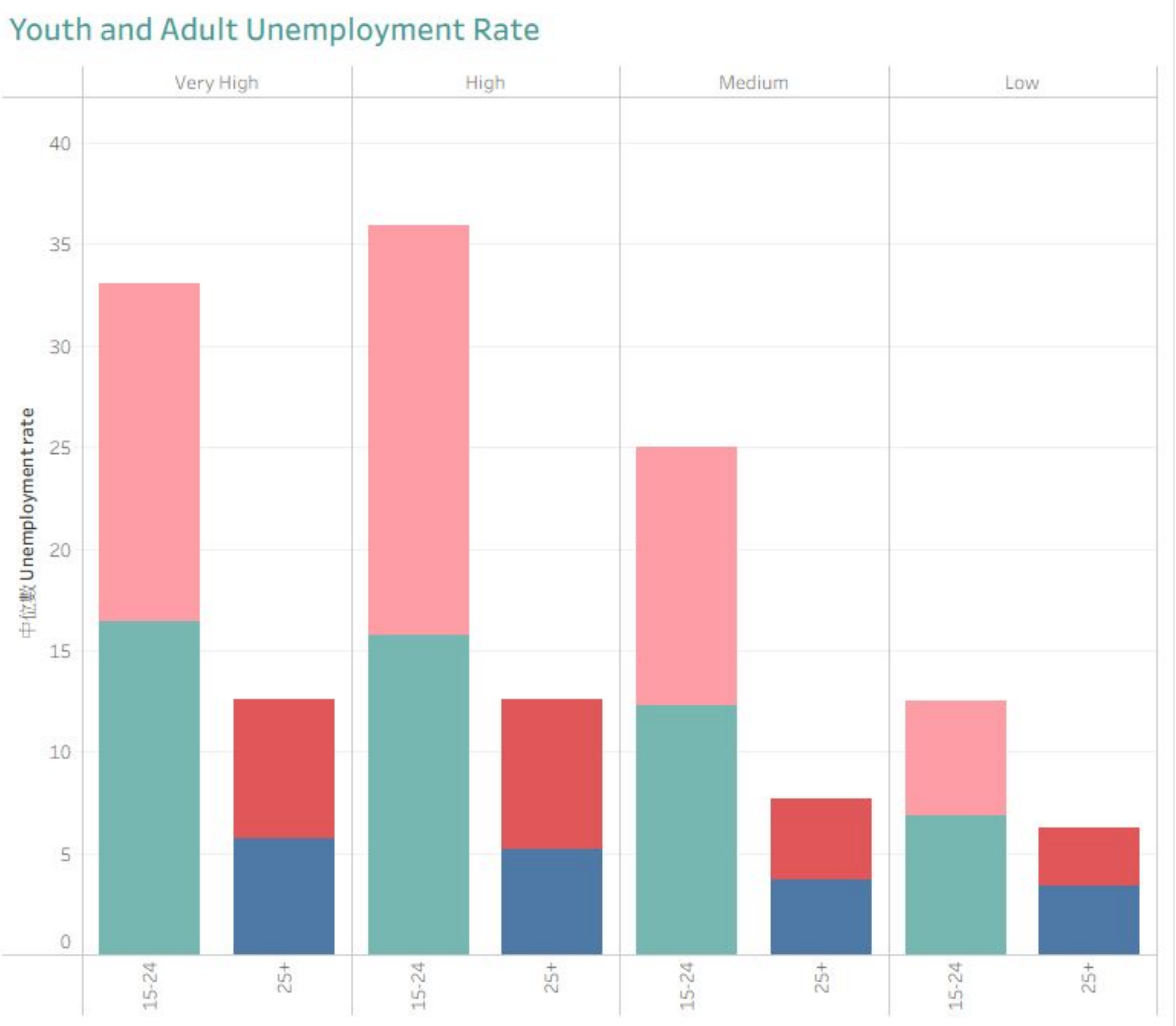
➔ Higher education level, larger entrance barrier
➔ Lower HDI, lower legal age to work

Highlights Age Inequality



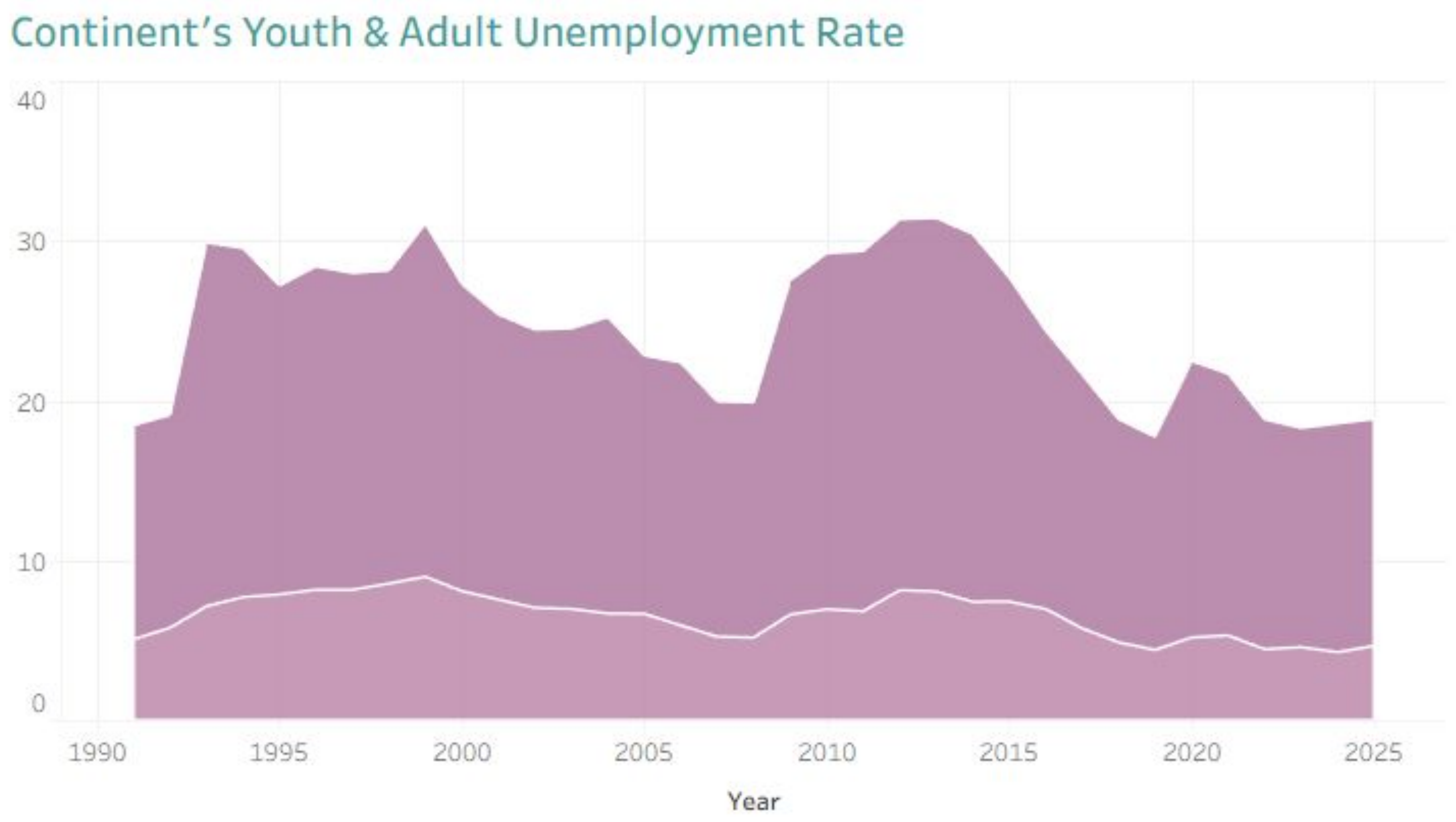
Even unemployment gap between the youth and adult

- ~ three times in difference

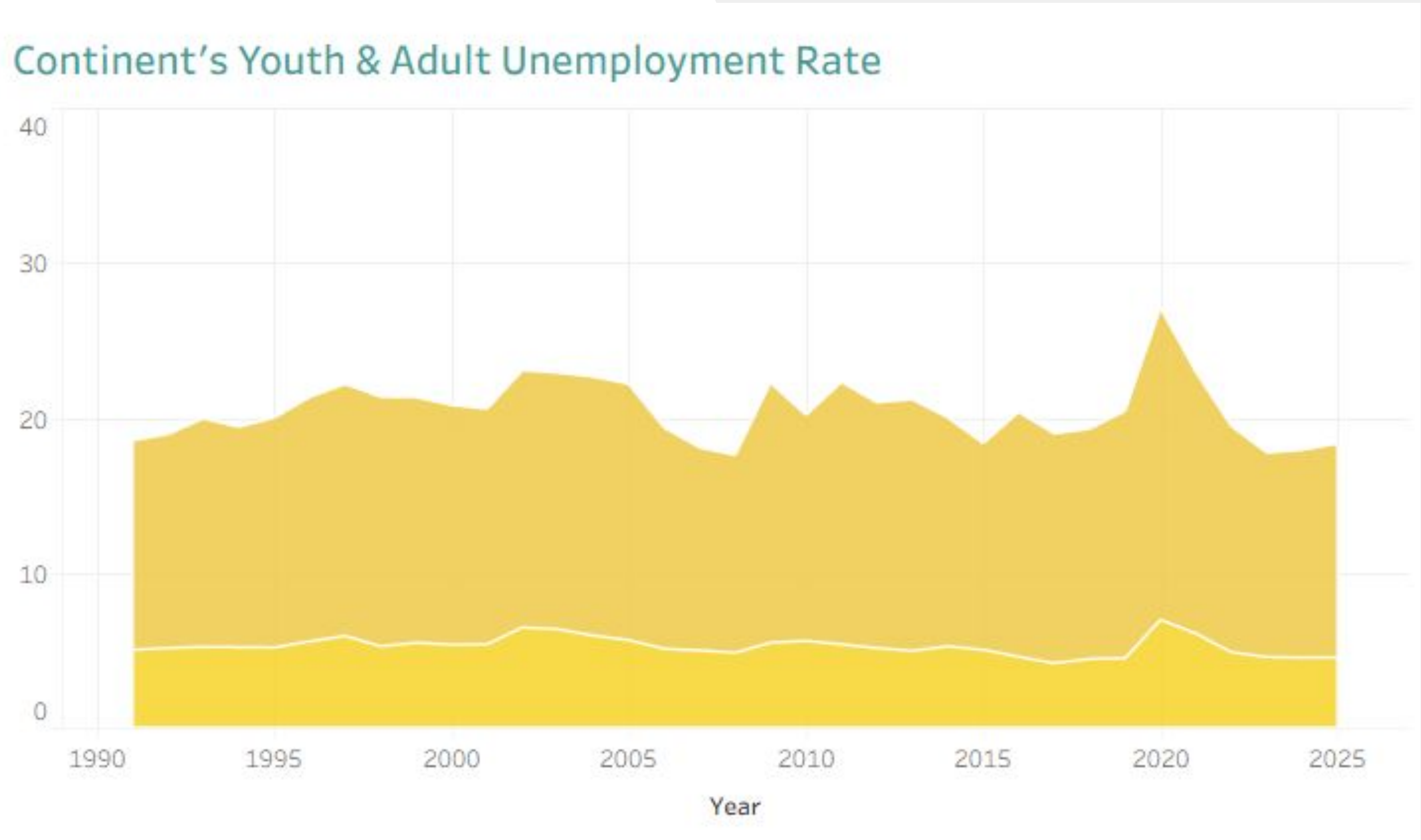


- Large gap between the youth and adult for “Very High” and “Highly developed” countries
- Smaller gap between the youth and adult for “Low” developed countries

Highlights Age Inequality



Among all continents and age groups, the unemployment rate of European youth is the highest



The unemployment rate of European and American adults with year is more fluctuated, compared to those in other continents

Conclusion

Global Unemployment Trend

Cyclical volatility

- Financial crisis and Pandemic
- Highly developed countries are more fragile

Gender Inequality

Highly developed

- Smaller Gender Unemployment gap because of feminism

Low to moderate developed

- Higher female unemployment rate because of sexism

Very low developed

- Lower female unemployment rate that female may do black-market jobs

Age Inequality

Highly developed

- Higher education barrier, higher unemployment gap between youth and adult

Low developed

- Lower legal age to work, lower unemployment gap between youth and adult

Challenges & Future Work

Challenges	Solutions
The original dataset only consists of 5 features ● Not enough for data analysis	Find other datasets which can join with the original one
Line graph plotted for different countries ● Numerous countries: difficult to visualize	Countries were clustered into continents for easy analysis and visualizations
Male & Female population is needed for more analysis on unemployment rate ● No such datasets	Simply take the median of unemployment rate for analysis

Challenges & Future Work

Things implemented in this project for simplicity	Future Work
• The Human Development Index (HDI) dataset for the year 2023 was combined with the original unemployment dataset which spans across 1991-2025	• A dataset which shows the HDI of each country for the past years should be found • More interesting analysis can be performed such as how the change in the HDI of a country affects its unemployment rate
• Leave the missing data blank for Human Development Index (HDI) and Education Level	• Machine learning algorithms like Random Forest and K-Nearest Neighbors (KNN) can be used to fill the missing data

Distribution of Work

Cheung Man Ting, Sabrina	Sze Hiu Laam, Cindy	Yam Ching Long, Angus
● Dataset Sourcing (Edu Level) ● Data Characteristics ● Dashboard Presentation	● Dataset Sourcing (HDI) ● Graph Plotting ● Insights & Conclusion	● Dataset Sourcing (Main data) ● Data Cleansing ● Insights

Thank You | Q&A

Main Dataset

Global Employment Unemployment Rates 1991-2025

https://www.kaggle.com/datasets/lucalullo/global-employment-unemployment-rates-1991-2025

Supplementary Dataset

Country Mapping

https://www.kaggle.com/datasets/andradaolteanu/country-mapping-iso-continent-region

Human Development Index (HDI) by Country

https://worldpopulationreview.com/country-rankings/hdi-by-country

Countries Education Level and Income

https://www.kaggle.com/datasets/alikarbala/countries-education-level-and-income